

Polynomial regression: BT2024060

Pulkit Pandey · personalised geothermal datasets · model selection and validation

1. Objective and supplied data

Predict the continuous target y for two separate problems using polynomial regression. var1 maps six plant settings to Net Power Score; var2 maps three spatial coordinates to Thermal Anomaly Score. Only this roll number's datasets were used.

Problem	Training	Test	Inputs / target
var1	1,000 labelled rows	1,000 unlabelled rows	x_1 - x_6 / y
var2	1,000 labelled rows	1,000 unlabelled rows	x_1 - x_3 / y

Inputs were already cleaned and bounded in $[-1, 1]$. Numeric and finite-value checks found no missing values. No observations were discarded or targets clipped.

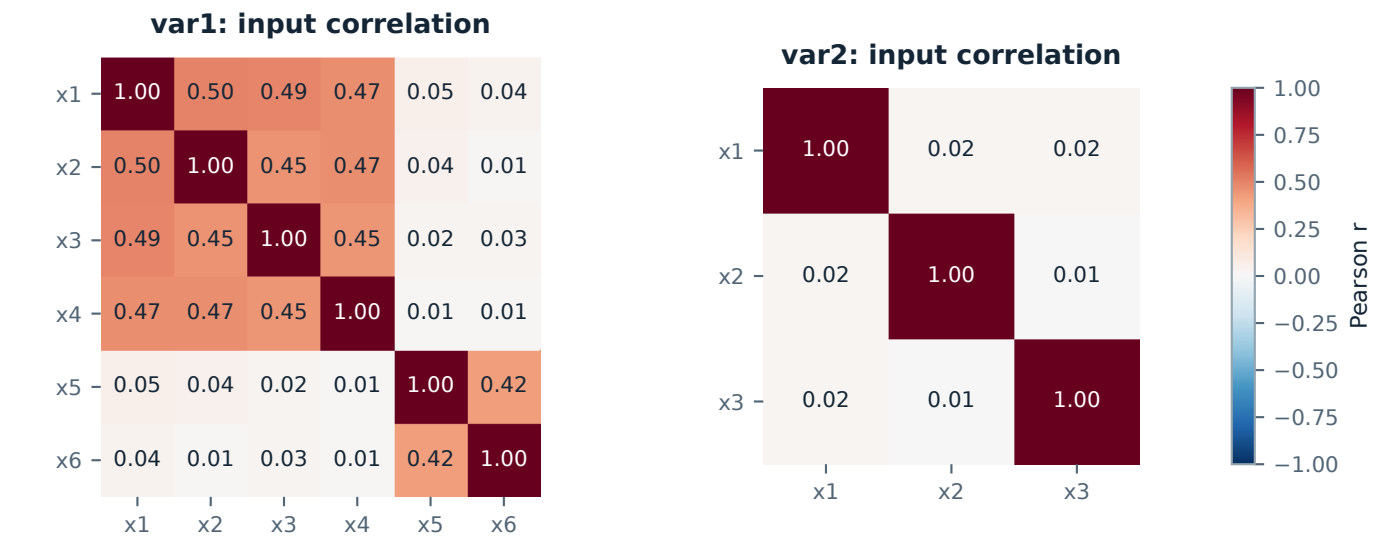


Figure 1. var1 has correlated groups (x_1 - x_4 and x_5 - x_6); var2 has weak pairwise linear correlations. Neither input correlation nor correlation with y alone establishes feature usefulness: nonlinear and interaction effects require validation.

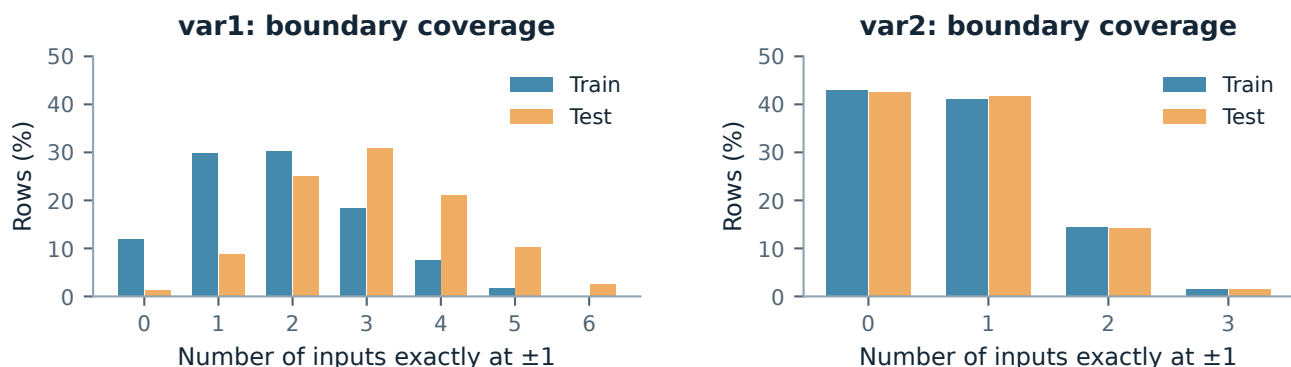


Figure 2. var1 test data are more boundary-heavy: 64.8% of test rows have at least three boundary inputs, versus 28.0% of all training rows. var2 input coverage is similar between train and test.

Repeated input rows were retained. var2 contains nine repeated training feature rows, with different y values at some identical coordinates. Identical inputs were kept together in every fitting/validation split to prevent group leakage.

Validation design and model selection

A fixed grouped split reserved 200 var1 and 202 var2 rows. Initial searches used five folds on the remaining 800 / 798 development rows. Polynomial expansion, feature scaling and fitting were applied within each fold; no validation targets entered fitting.

$$\hat{y} = \beta_0 + \sum_{1 \leq a_1 + \dots + a_m \leq d} \beta_a \prod_{j=1}^m x_j^{a_j}$$

Total degree d includes cross-feature interactions. Expanded terms were standardised before Ridge (L2) or Lasso (L1) penalties; their α values are not directly comparable.

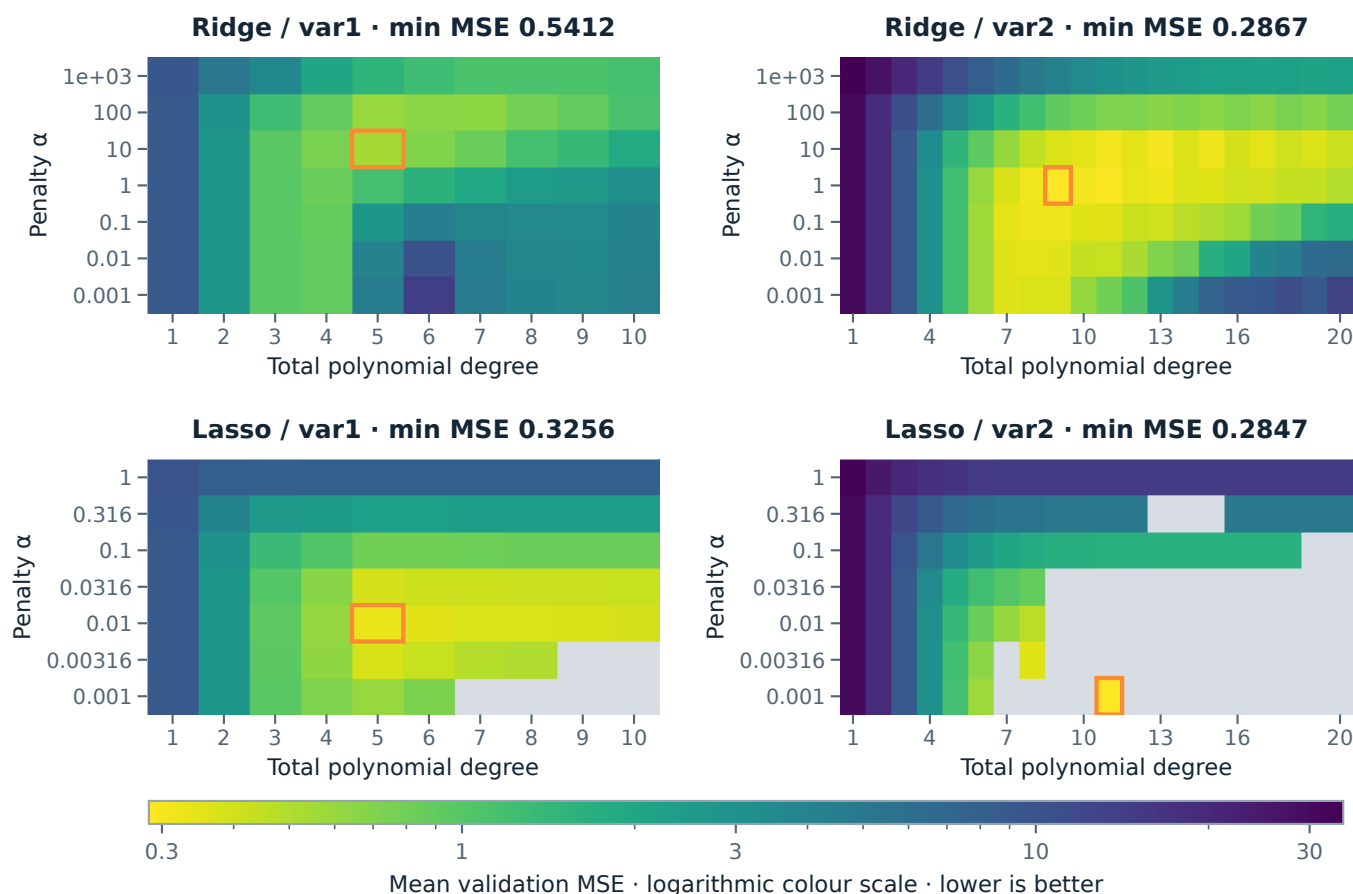


Figure 3. Full degree ranges: var1 1-10; var2 1-20. Outlines identify the minimum on each plotted grid. Grey cells failed convergence in at least one fold. The var2 Lasso degree-11, $\alpha=0.001$ cell includes its successful longer verification run.

Refinement beyond the base grids

Problem	Ridge base grid	Ridge refined	Lasso converged
var1	d=5, $\alpha=10$ MSE 0.5412	d=5, $\alpha=30$ MSE 0.5012	d=5, $\alpha=0.01$ MSE 0.3256
var2	d=9, $\alpha=1$ MSE 0.2867	d=11, $\alpha=3$ MSE 0.2789	d=11, $\alpha=0.001$ MSE 0.2847

Table 1. These are mean MSE values on the original development folds. Fine searches between the base penalties explain why the final Ridge degree differs from its grid winner.

Screening all 63 var1 and seven var2 feature subsets near leading degrees favoured all inputs. Six inputs at degree 10 produce 8,007 terms; regularisation and validation control complexity. The search was staged, not exhaustive.

Final choices and robustness checks

All supplied features were retained. Final choices were fixed before scoring the reserved holdout; the submitted polynomials were subsequently refitted on all training rows.

var1 · degree 5 polynomial

50% Lasso ($\alpha=0.01$) + 50% OLS refit of terms selected by Lasso ($\alpha=0.03$). The equal average remains a degree-5 polynomial.

var2 · degree 11 polynomial

Ridge $\alpha=3$ on all three inputs. The expansion contains 363 terms plus an intercept. All fitting steps stay inside validation folds.

Additional techniques and fresh-split confirmation

255 configurations/control settings compared Elastic Net, Legendre/Chebyshev bases, sparse coefficient refits and boundary weighting. Alternative bases did not improve the leaders; var1 boundary weighting worsened errors. Six shortlisted candidates per problem were tested on five fresh shuffled 5-fold group splits (25 fits each).

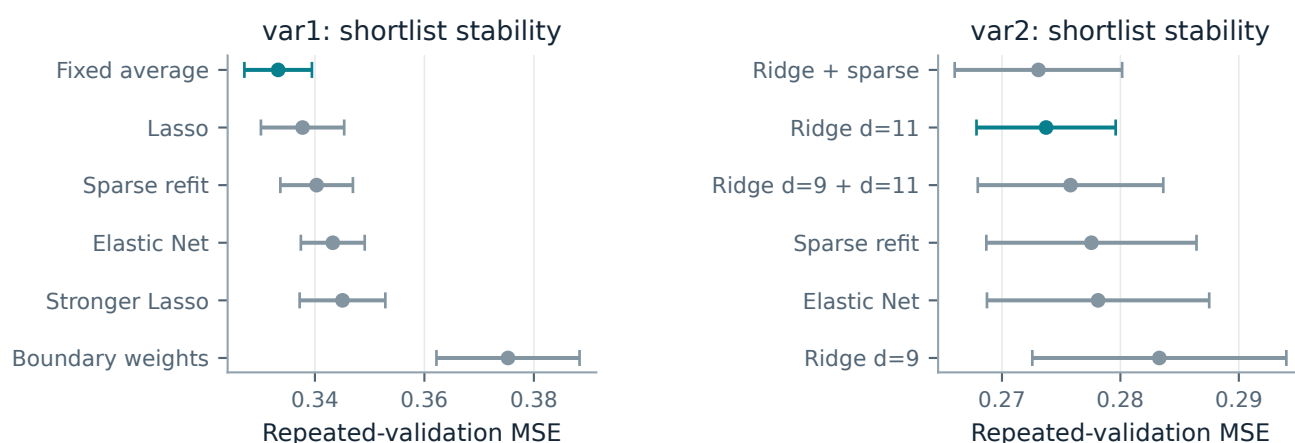


Figure 4. Points are mean MSE; bars are SD across five split repetitions, not confidence intervals. The var1 average improved all five repetitions ($0.3377 \rightarrow 0.3333$). var2's sparse refit gain was inconsistent; Ridge was retained for its stability and lower worst-split MSE.

Learning curves at fixed settings

Each curve uses five validation folds and progressively more fitting rows. Every selected-model fit converged. Degree and penalties were chosen through validation.

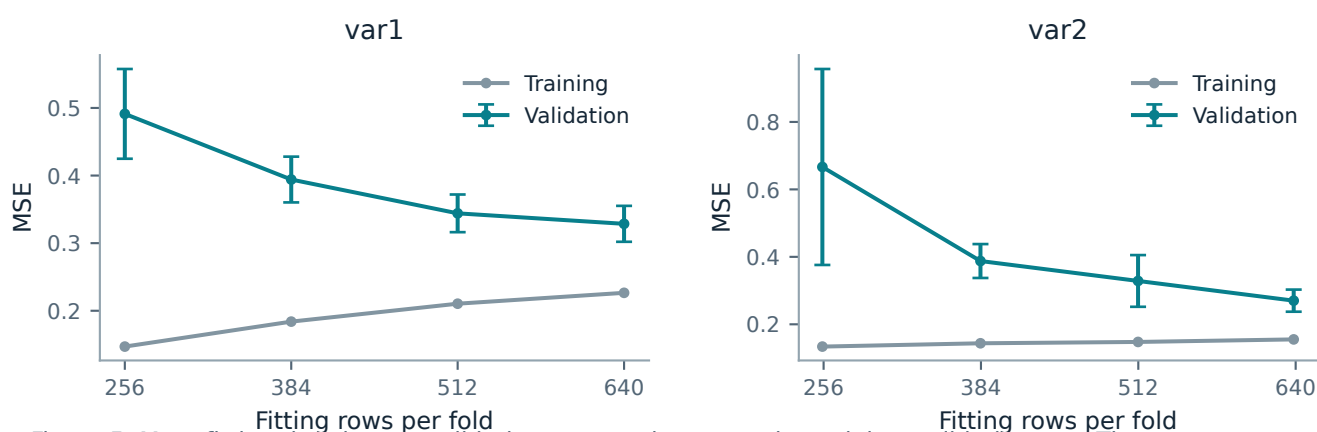


Figure 5. More fitting data lowers validation error and narrows the training-validation gap. The curves support using all 1,000 rows for the final fit; they do not prove zero overfitting.

Reserved-holdout results and limitations

Selected configurations and training-file hashes were saved before assessment. The holdout was excluded from fitting and tuning, and its scores did not change the choices. Initial descriptive EDA covered all supplied training rows. Hidden test labels remain unavailable.

Problem	CV MSE	CV R ²	Holdout n	Holdout MSE	Holdout R ²
var1	0.3333	0.9668	200	0.3184	0.9676
var2	0.2737	0.9933	202	0.3982	0.9934

Table 2. CV metrics are averages over complete out-of-fold predictions from five repetitions. Holdout metrics use one fixed model fit on the development partition.

Approximate 95% group-bootstrap intervals (2,000 resamples):

var1: MSE [0.2616, 0.3769], R² [0.9570, 0.9748].

var2: MSE [0.2489, 0.6194], R² [0.9911, 0.9950].

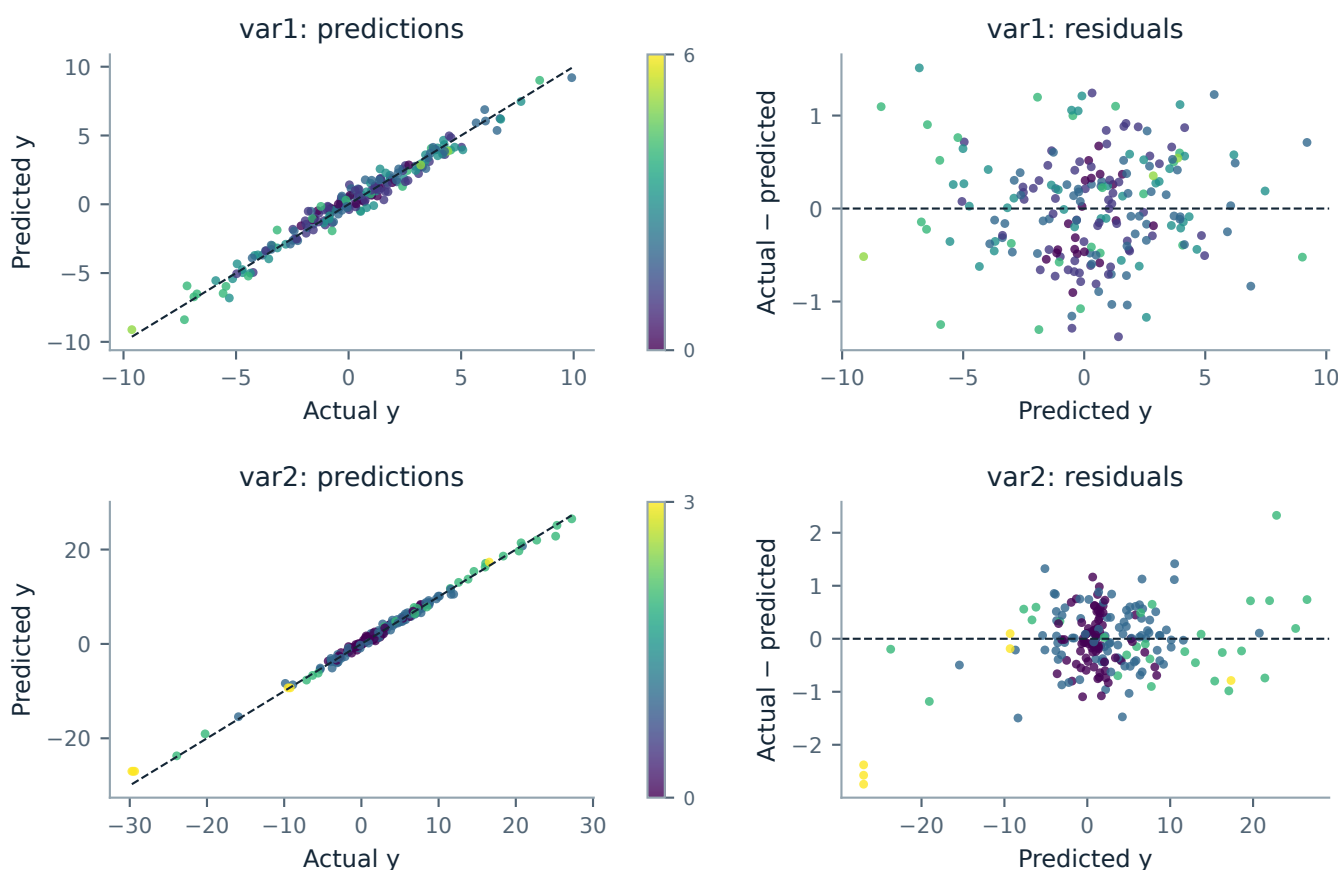


Figure 6. Colour indicates the number of inputs exactly at ± 1 . Mean holdout residuals are $+0.0062$ (var1) and -0.0403 (var2). var2's six full-corner observations have MSE 3.4145, versus 0.2341 for 77 interior rows; a repeated corner accounts for several large errors.

Interpretation and practical limits

The results support generalisation within the sampled domain, but cannot establish globally optimal parameters or zero overfitting/underfitting. var1's test boundary shift and var2's rare corner groups limit certainty at unobserved combinations. Repeated splits share observations; their SD measures stability. The holdout, not further tuning to its scores, provides the final assessment.

Rationale, reproduction and deliverables

Why these polynomial degrees and techniques?

var1: degree-1 regression had development MSE 8.9374. Ordinary degree 5 fitted training data closely (MSE 0.0733) but generalised poorly (validation MSE 4.5667). Lasso $\alpha=0.01$ reduced the original-fold validation MSE to 0.3256. The fixed average with a sparse OLS refit further improved fresh-split consistency while remaining degree 5.

var2: degree-1 MSE was 30.8325; degree 4 gave 3.0693. Higher-degree regularised models captured the nonlinear structure, and degree-11 Ridge $\alpha=3$ remained consistent across fresh splits. Adding complexity beyond this region did not reliably improve validation. Degrees and penalties were justified jointly using validation, convergence checks and learning curves. Final selection considered both error and stability.

Within the three repeated-input groups in var2 development data, pooled target variance was about 0.2253 (six residual degrees of freedom). This suggests noise, but is too limited to establish a global noise floor. Weak individual feature correlations were not used to remove inputs.

Final fitting and prediction format

After assessment, the fixed models were trained on all 1,000 labelled rows for each problem. Predictions preserve test row order. Repeated inputs receive identical values; all predictions were checked to be finite. The sample format contains one y column and no index column.

Prediction file	Rows	Columns
BT2024060_pred_var1.csv	1,000	y only
BT2024060_pred_var2.csv	1,000	y only

Code and reproducibility

Training, inference and analysis code are organised under scripts/ in the repository below. requirements.txt pins the Python packages. Saved configurations, original row splits, metrics and diagnostic figures are available under outputs/.

```
python -m pip install -r requirements.txt
python scripts/final_model_check.py
python scripts/predict_submission.py
```

final_model_check.py refits the frozen choices and reuses the saved holdout assessment. predict_submission.py reproduces both CSVs from exported coefficients without retraining. JSON/CSV coefficient exports reproduce pipeline predictions to approximately 10^{-14} , including the combined var1 polynomial.

GitHub repository

<https://github.com/COolAlien35/polynomial-regression>

Submission artifacts in submission/: this five-page PDF and the two prediction CSVs; also submit the repository containing training and inference code. Reported scores are internal validation and reserved-holdout scores; the assignment's hidden-test scores are unknown.